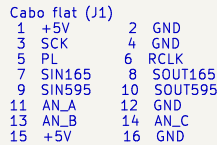


Etiqueta + 16 entradas + 16 saídas para LED. Monte sempre todos os CIs:
a posição de cada byte na cadeia depende disso. Entrada ou saída sem uso fica livre.



Etiqueta: chave k ligada (ON) = bit k-1 igual a 1.
Bits 7-6 = tamanho (médio: 10); bits 5-0 = número da seção (1 a 62).

Entradas: SIN165 > U3 > U2 > U1 > SOUT165 (para o Mega).
O Mega lê nesta ordem: U1 (etiqueta), U2 (IN0-7), U3 (IN8-15).
Botão ou chave: entre INN e GND. Apertado/ligado = 0 (pull-up de 10k).

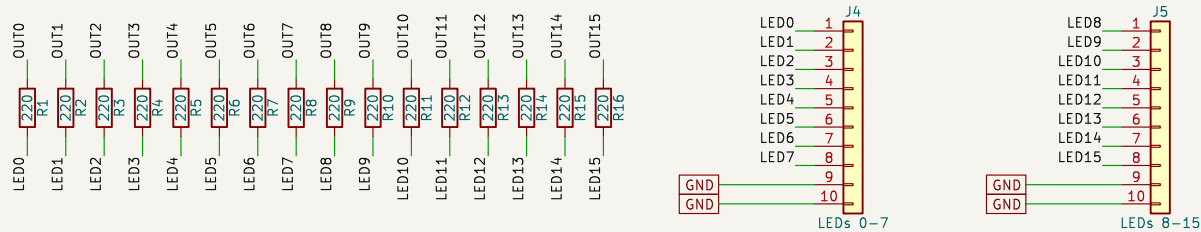
Saídas: SIN595 > U4 > U5 > SOUT595 (para o próximo slot).
LED: anodo em LEDn, catodo no GND (220: uns 13 mA por LED;
evite os 8 LEDs de um 74HC595 acesos juntos: passa de 70 mA).
Quem acende os LEDs é o Mega, com o estado que vem do jogo:
nenhum LED é ligado direto a um botão.

The image contains two circuit diagrams for connecting a 74HC595 shift register to an Arduino Uno. Both diagrams show the shift register's VCC connected to +5V and GND connected to GND. The input pins are connected as follows:

- Left Diagram (74HC595):**
 - SER (pin 14) is connected to Arduino pin 14 (SIN595).
 - SRCLK (pin 11) is connected to Arduino pin 11 (SCK).
 - SRCLR (pin 10) is connected to Arduino pin 10 (+5V).
 - RCLK (pin 12) is connected to Arduino pin 12 (RCLK).
 - OE (pin 13) is connected to Arduino pin 13 (GND).
 - Pin 8 is connected to GND.
- Right Diagram (CH595_1):**
 - SER (pin 14) is connected to Arduino pin 14 (CH595_1).
 - SRCLK (pin 11) is connected to Arduino pin 11 (SCK).
 - SRCLR (pin 10) is connected to Arduino pin 10 (+5V).
 - RCLK (pin 12) is connected to Arduino pin 12 (RCLK).
 - OE (pin 13) is connected to Arduino pin 13 (GND).
 - Pin 8 is connected to GND.

The output pins (QA-QH) are connected to LEDs. The left diagram shows outputs QA-QH connected to pins 15-21, and the right diagram shows outputs QA-QH connected to pins 15-21.

Um 100nF colado em cada CI



Liga no backplane por um cabo flat de 16 vias (IDC 2x8)
Etiqueta + 16 entradas (2 x 74HC165) + 16 saídas para LED (2 x 74HC595)

KSP Cockpit

Sheet: /
File: modulo_medio.kicad_sch

Title: Módulo médio do painel

Size: A3	Date: 2026-09-26
----------	------------------

KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4
--------------	-------------------------------

7

Rev: 1

Id: 1/1